The Rest Pose: why MABEL's zero configuration is not MABEL at rest, and the gravity-equilibrium posture that replaces it

MABEL project

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Abstract

Every client of the MABEL stack — the browser choreography studio, the iPhone and Vision Pro applications, the offline demonstration viewer — built its default whole-body pose by filling a joint dictionary with zeros. That is the natural thing to write and it is wrong. The robot's zero configuration is not the robot at rest: at $q = 0$ the upper arm hangs but the forearm projects horizontally forward, so the two segments meet at 91.3°, and the two arms together demand 38.587 N m of static gravity holding torque to maintain. The arm only hangs when the elbow is *extended*, which on this kinematic chain is $q_4 = \pm 1.52$ rad, not 0. We derive the minimum-holding-torque rest posture directly from the MuJoCo model by minimising $\sum \tau^2$ under inverse dynamics at $\dot{q} = \ddot{q} = 0$, obtaining 1.638 N m — a 24-fold reduction — with the hands 0.369 m below the shoulders and 0.148 m outboard of them, clear of both torso and thighs. The residual 15.1° of elbow bend is shown to be the linkage's mechanical floor rather than a tuning choice. Two defects in the shipped model surfaced during the derivation and are reported here: the two elbows carry non-mirrored joint limits, so the previously shipped left-arm rest value lay 0.038 rad outside its own limit, and the lift column self-intersects the chassis below 0.155 m per stage, leaving only the upper half of its declared stroke usable in simulation.

Keywords: rest pose, gravity equilibrium, static holding torque, inverse dynamics, joint limits, kinematic calibration, model verification.

1 Overview

A humanoid's *rest pose* is the configuration it occupies when nothing is asking it to be anywhere: the pose a viewer sees before a motion starts, the pose a clip returns to when it ends, the pose the robot glides to on `Home`. It is therefore the single most frequently displayed and most frequently commanded configuration in the system, and the one whose cost is paid continuously rather than transiently.

The operator report that prompted this work was visual — *"the default beginning and ending pose… the elbow should be extended down, at the naturally gravity down position"* — but the underlying defect is mechanical, and it is a defect of *convention*, not of any one client. MABEL's joint zero for the elbow is not the extended arm; it is the arm folded to a right angle. A dictionary of zeros is consequently a command to stand with both forearms projecting forward, which every client duly rendered and, on the real robot, duly held.

This document derives the replacement posture from the model (§2), reports what it costs against the two conventions it replaces (§3), states where the value must be kept in sync (§4), and

records two model defects found on the way (§5). The whole-body stability model that consumes the rest pose is documented separately in `controller/docs/wbc_stability.tex`; the choreography library that starts and ends every clip here is `web_gui/control_center/docs/motion_library.tex`.

2 Method

2.1 Static holding torque

Let $q \in \mathbb{R}^n$ be the full configuration and $M(q)\ddot{q} + C(q,\dot{q}) + g(q) = \tau$ the manipulator equation. At rest, $\dot{q} = \ddot{q} = 0$, so the actuator torque required to hold the configuration is exactly the gravity term,

$$\tau_{\text{hold}}(q) \;=\; g(q), \tag{1}$$

which we evaluate with `mj_inverse` on the canonical model, `mabel_full.xml`, after seeding the assembled `home` keyframe and zeroing $\dot{q}$ and $\ddot{q}$. Writing $\mathcal{A}$ for the fourteen arm joints, the cost to be minimised is

$$J(q) \;=\; \sum_{i \in \mathcal{A}} \tau_{\text{hold},i}(q)^2. \tag{2}$$

This is the right objective for a rest pose specifically: it is the power the machine burns doing nothing, the torque the servos must hold against indefinitely, and — because a position servo answers a persistent error with a persistent effort — the pose's thermal cost.

2.2 Search space, and why it is only two dimensions

The arms are constrained to mirror, $q^{\text{right}} = -q^{\text{left}}$, which the model's joint-limit tables respect (with one exception, §5). Of the seven joints per arm, three are rolls about the limb axis (q_3, q_5, q_7) and one is a wrist pitch (q_6); none of them changes the gravity torque of a hanging arm to first order, and all are left at zero so the forearm and hand sit level. The shoulder pitch q_1 is likewise pinned at 0 — not because J is insensitive to it (it is not; the unconstrained optimum is $q_1 = 0.0126$) but because every clip in the motion library is authored as an amplitude on q_1 measured from zero, and moving the origin by $0.7°$ would silently redefine all of them for a $0.7\,\text{N m}$ gain. What remains is

$$\min_{q_2,\,q_4} J(q_2, q_4) \quad \text{subject to} \quad q_2 \in [\,\underline{q}_2, \overline{q}_2\,],\;\; q_4 \in [\,\underline{q}_4, \overline{q}_4\,], \tag{3}$$

that is, the shoulder roll — how far the arm is held off the body — against the elbow, with everything else fixed by the argument above.

2.3 The elbow margin

J is monotone decreasing in q_4 over the whole feasible interval: the arm always wants to be straighter, so the minimiser of (3) sits exactly on the elbow's upper limit, $\overline{q}_4 = 1.542\,\text{rad}$. A rest pose that lives on a hard stop is a rest pose that grinds, so the shipped value backs off by

$$q_4^\star \;=\; \overline{q}_4 - \delta, \qquad \delta = 0.022\,\text{rad}, \tag{4}$$

and $q_2^\star$ is then re-minimised at that fixed $q_4^\star$. The margin costs $0.715\,\text{N m}$ ($0.923\,\text{N m}$ at the stop versus $1.638\,\text{N m}$ at $q_4^\star$), which buys not pressing a $27\,\text{N m}$ actuator against a limit for the robot's entire idle life. Both arms use the same δ, so the pose stays exactly mirrored even though their limits are not.

2.4 Reference-carrying joints

The wire commands *absolute* configuration, not displacement from the model's reference. Four joints have a non-zero `ref` in the MJCF — the two ORCA wrists at ± 0.15 rad and the two lift stages at 0.158 75 m — so their rest value is that reference and commanding 0 would twist each wrist by $-$ref and drive the lift to the floor. This is the same absolute/relative distinction that governs the three-dimensional clients, which render $q - \text{ref}$; it is documented once in the repository root guidance and is restated here because the rest pose is where it bites.

3 Experiments

All numbers below are produced by `controller/experiments/exp_rest_pose.py`, which regenerates Figure 1 from the model in a single pass. No value in this section is hand-entered.

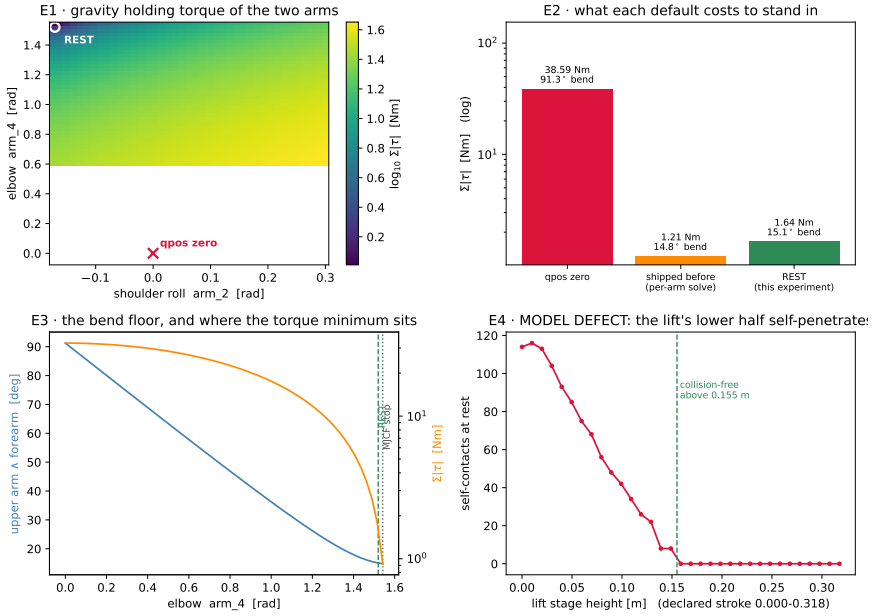


Figure 1: **E1** (top left) the static holding torque of the two arms over the shoulder-roll/elbow plane, $\log_{10} \sum |\tau|$; the cross marks the zero configuration every client used as its default and the circle marks the derived rest pose. **E2** (top right) what each candidate default costs to stand in, log scale. **E3** (bottom left) elbow bend and holding torque as functions of the elbow joint: the bend floor is mechanical and the torque minimum sits on the joint stop. **E4** (bottom right) the model defect: self-contacts against lift height.

3.1 E1 — the landscape

Sweeping 41×41 points over $q_2 \in [-0.30, 0.30]$ and $q_4 \in [0.6, \overline{q}_4]$ spans 4 orders of magnitude in $\sum |\tau|$, from 1.024 N m at the minimum to well over 30 N m in the folded-elbow corner where $q = 0$ lies. The landscape is smooth and has a single basin, so the two-parameter reduction of §2 is not hiding a better solution elsewhere.

3.2 E2 — against the two conventions it replaces

Table 1: Candidate rest poses, measured on the canonical model. "Bend" is the angle between upper arm and forearm (180° would be perfectly straight; see §5 for why 15° is the floor here). "Drop" is the hand's height below its shoulder.

Pose	$\sum \lvert\tau\rvert$ [N m]	$\max \lvert\tau\rvert$ [N m]	bend [°]	drop [m]	in limits
$q = 0$ (every client's default)	38.587	7.108	91.3	0.152	yes
per-arm solve (shipped before)	1.209	0.270	14.8	0.370	**no**
REST (this work)	1.638	0.382	15.1	0.369	yes

Table 1 gives the comparison. The zero configuration costs 24 times the rest pose in total holding torque and 19 times in the worst single joint, and it places the hands 0.216 m in front of the shoulders rather than 0.013 m, which is the visual difference the operator reported. The previously shipped posture is close in torque — it was derived from the same principle, per arm — but it is *infeasible*: its left elbow value of 1.5795 rad lies 0.038 rad outside that joint's declared limit. MuJoCo clamps silently, so the simulation looked correct; the physical actuator would have pushed into its stop. Its per-arm derivation also left the two arms 0.002 rad to 0.008 rad apart on every joint, so the robot was never quite commanded a symmetric pose.

3.3 E3 — the bend floor is mechanical

Sweeping the elbow across its feasible interval, with q_2 re-minimised at each point, the upper-arm/forearm angle bottoms out at 14.8° exactly at $\overline{q}_4$. The arm cannot be made straighter than that because the forearm's link frame is not collinear with the upper arm's at the elbow's end-stop: 15° is the geometry of the linkage, not a residual of the optimisation. Over the same sweep $\sum \lvert\tau\rvert$ falls monotonically, confirming the end-stop argument of §2.

3.4 E4 — the lift's usable stroke

Holding the rest pose and sweeping the lift, the model reports zero self-contacts at and above 0.155 m per stage, 7 contacts at 0.150 m, and 116 when fully retracted. The declared stroke is $[0, 0.3175\,\text{m}]$ per stage; only its upper half is usable in simulation. The rest pose sits at the assembled reference, 0.158 75 m, which is 0.4 mm above that boundary — the model can barely represent its own rest configuration, and cannot represent a retracted one at all.

4 Implementation and usage

The pose is

$$q_1 = 0, \quad q_2 = \mp 0.1711, \quad q_3 = 0, \quad q_4 = \pm 1.52, \quad q_{5,6,7} = 0,$$

(left sign first), with the wrists at ±0.15, both lift stages at 0.158 75, torso and neck at zero, the steer joints at zero, and a relaxed grip of 0.2. It is replicated in exactly four places, and they must agree:

- `web_gui/control_center/web/js/motions.js` — `REST`, the frame every choreography clip starts and ends in;
- `web_gui/control_center/web/js/homepose.js` — `REST_POSE`, the target of the studio's `Home` glide (adds the steer joints);

- `Services/RobotController.swift` — `restBodyPose` plus `homeGrip`, in both the iPhone and Vision Pro applications;
- `Models/JointModel.swift` — `RobotModel.armsAtRest`, which `homePose` composes with the reference-carrying joints, in both applications.

The agreement is gated rather than trusted. `gen_motion_clips.py` records `REST` into the baked table the applications ship, and `MotionClipTests.testTheAppsHomePoseIsTheStudiosRestPose` fails if `RobotModel.homePose` diverges from it; `tests/js/homepose.test.mjs` checks every joint of `REST_POSE` against the model's own limits. To re-derive after a model change:

```
python3 controller/experiments/exp_rest_pose.py      # numbers + figure
```

5 Limitations

The elbows are not mirrored in the model. `Right Arm 4` declares $[-1.670, 0.773]$ while `Left Arm 4` declares $[-0.943, 1.542]$; the mirror of the right would be $[-0.773, 1.670]$. The left arm therefore extends 0.128 rad (7.3°) less than the right in the model. We have treated the MJCF as authoritative and backed off the tighter of the two symmetrically, so the shipped pose is legal under either reading, but one of the two numbers is a transcription error and should be resolved against the hardware. Until it is, the true minimum-torque rest pose is unknown to within that margin.

The retracted lift is not representable. §3 E4. The consequence for the motion library is that expressive clips must express "lower" as travel above 0.155 m rather than toward zero; the one primitive that genuinely asks for zero, `lift_down`, is left alone, because clamping it would remove a real capability of the real column to work around a collision-geometry error. This should be fixed in the MJCF.

The objective is static. J in (2) scores holding torque, not reachability or time-to-task. A rest pose with the arms hanging is farther from a forward reach than one with the elbows pre-folded, so it costs a fraction of a second of transit on every task that starts from rest. We consider that the right trade for a pose held indefinitely, but it is a trade and it has not been measured.

No hardware verification. Every number here is from the model. The holding torques predicted for $q = 0$ (7.1 N m on a single elbow) are large enough to be measurable on the real arm, and that measurement has not been made.

References

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